Vieta's Formulas

1. The roots of $2x^2 + 2x - 3 = 0$ are r and s. Find $r + s$ and rs.

2. A quadratic $x^2 + bx + c = 0$ has roots r and $2r$. Express c in terms of b.

3. The two roots of $x^2 - px + q = 0$ are both equal. Express q in terms of p.

4. The roots of $x^2 + bx + c = 0$ are p and q. If the lines $y = px + 3$ and $y = qx - 3$ are perpendicular, what is c?

5. The roots of $7x^2 - 6x + 7 = 0$ are r and s. Compute $r^2 + s^2$.

6. The roots of $2x^2 - 7x + 3 = 0$ are r and s. Compute $\frac{1}{r} + \frac{1}{s}$.

7. The roots of $3x^2 - 6x + 2 = 0$ are r and s. Compute $\frac{r}{s} + \frac{s}{r}$.

8. The roots of $x^2 - 5x + 3 = 0$ are r and s. Find the monic quadratic whose roots are $r + 1$ and $s + 1$.

9. The roots of $x^2 - 4x + 1 = 0$ are r and s. Find the monic quadratic whose roots are $2r$ and $2s$.

10. The roots of $5x^2 - 4x + 2 = 0$ are r and s. Find the monic quadratic whose roots are $r^2 - 2$ and $s^2 - 2$.

11. The roots of $4x^2 - 3x + 1 = 0$ are r and s. Find the monic quadratic whose roots are $r + \frac{1}{s}$ and $s + \frac{1}{r}$.

12. The roots of $x^2 - 5x + 1 = 0$ are r and s. Find the monic quadratic whose roots are $r^2 + r$ and $s^2 + s$.

13. The roots of $x^2 - 4x + 12 = 0$ are r and s. Compute $r^3 + s^3$.

14. The roots of $x^2 - 5x + 3 = 0$ are r and s. Compute $r^4 + s^4$.

15. The roots of $x^2 - 2x - 1 = 0$ are r and s. Compute $r^5 + s^5$.

16. The roots of $x^2 - x - 1 = 0$ are ϕ and $\hat{\phi}$ (the golden ratio and its conjugate). Compute $\phi^6 + \hat{\phi}^6$.

17. The roots of $x^2 - 4x + 2 = 0$ are r and s. Find a recurrence relationship for $r^n + s^n$ and use it to compute $r^{10} + s^{10}$.

18. The roots of $2x^3 + 6x^2 + 11x + 8 = 0$ are r, s, t. Find $r + s + t$, $rs + rt + st$, and rst.

19. The roots of $x^3 - 7x + 5 = 0$ are r, s, t. Find $rs + rt + st$.

20. The three roots of $x^3 - 9x^2 + 26x - 24 = 0$ are consecutive integers. Find the roots.

21. A cubic $x^3 + px^2 + qx + r = 0$ has roots $1, 2, 3$. Find p, q, and r.

22. The roots of $3x^3 - 6x^2 + 8x - 6 = 0$ are r, s, t. Compute $r^2 + s^2 + t^2$.

23. The roots of $x^3 - 4x^2 + 5x - 2 = 0$ are r, s, t. Compute $\dfrac{1}{r} + \dfrac{1}{s} + \dfrac{1}{t}$.

24. The roots of $x^3 - 3x^2 + 7x - 1 = 0$ are r, s, t. Without factoring, compute $r^3 + s^3 + t^3$.

25. Let r, s, t be the roots of $x^3 - 2x^2 - x + 2 = 0$. Compute $r^2s^2 + r^2t^2 + s^2t^2$.

26. Let r, s, t be roots of $x^3 - 5x^2 + 8x - 4 = 0$. Compute $\dfrac{1}{rs} + \dfrac{1}{rt} + \dfrac{1}{st}$.

27. The roots of a monic cubic are in arithmetic progression with common difference 2 and product 15. Find the cubic.

28. A monic quadratic with integer coefficients has a root $3 + \sqrt{5}$. Write it.

29. A monic cubic with rational coefficients has roots 1 and $2 + i$. Write it.

30. A monic quartic with rational coefficients has roots $\sqrt{2}$ and $1 + \sqrt{3}$. Write it.

31. The roots of $x^2 + bx + 12 = 0$ are both positive integers. Find all possible b.

32. The equation $x^2 + kx + 4k = 0$ has two equal roots. Find all possible values of k.

33. For what values of k does $x^2 - 2kx + k + 6 = 0$ have roots that are both positive?

34. The roots r and s of $x^2 - mx + n = 0$ satisfy $r^2 + s^2 = 7$ and $r^3 + s^3 = 10$. Find m and n.

35. The equation $x^2 + px + q = 0$ has roots r and s. The equation $x^2 + qx + p = 0$ has roots u and v. If r, s, u, v are all distinct, find the value of $r + s + u + v$.

36. A quadratic $x^2 - ax + b = 0$ has roots r and s with $r < s$. Given $r + s = 5$ and $r^2 + s^2 = 13$, find a and b.

37. The roots of $x^2 + px + q = 0$ are r and s, and the roots of $x^2 + qx + p = 0$ are r^2 and s^2. Find all possible values of $p + q$.

38. The polynomial $x^2 - (a + b)x + ab$ is given. A new polynomial is formed by replacing each root with its square. If the new polynomial is $x^2 - 13x + 36$, find all pairs (a, b) of integers.

39. The roots of $x^2 - 4x + 1 = 0$ are r and s with $r > s$. Compute $\sqrt{r} - \sqrt{s}$.

40. Let r_1, r_2, r_3, r_4 be the roots of $x^4 - 8x^3 + 14x^2 + 8x - 15 = 0$. Compute $r_1^2 + r_2^2 + r_3^2 + r_4^2$.

41. Let r_1, r_2, r_3, r_4 be the roots of $x^4 - 6x^2 + 1 = 0$. Compute $r_1^4 + r_2^4 + r_3^4 + r_4^4$.

42. Let r and s be roots of $x^2 - 63x + k = 0$ for some positive integer k. If r and s are both positive integers, find the number of possible values of k.

43. The roots of $x^3 - 3x^2 + 4x - 5 = 0$ are a, b, c. Compute $(a + b)(b + c)(a + c)$.

44. Let r_1, r_2, r_3 be the roots of $x^3 - 3x - 1 = 0$. Compute $r_1^3 + r_2^3 + r_3^3$.